Lecture 8

Def: The vector $\vec{e_j}$ is the vector whose j^{th} entry is 1 and the rest all zero

Thm: let $T: \mathbb{R}^n \to \mathbb{R}^m$

Then there is a unique matrix $A \in \mathbb{R}^{m \times n}$ such that

$$T(\vec{x}) = A\vec{x}$$

for all $\vec{x} \in \mathbb{R}^n$, satisfies:

$$A = [T(\vec{e_1}), T(\vec{e_2}), \cdots, T(\vec{e_n})]$$

column vector

Proof: According to def

$$\vec{e_j} = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad \vec{e_j} \in \mathbb{R}^n$$

the jth component

$A \in \mathbb{R}^{m \times n}$, set $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix}$

$\because$ for $\vec{e_j} \in \mathbb{R}^n$

$$T(\vec{e_j}) = A\vec{e_j} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

↑ The jth column vector of A

Some Geometric Examples of Linear Transformation in 2D

1. Reflection About the x-axis / y-axis

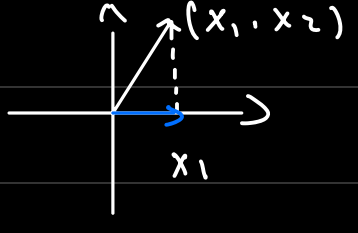
$$A = \begin{bmatrix} \pm 1 & 0 \\ 0 & \mp 1 \end{bmatrix}$$

2. Reflection Through the Origin

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

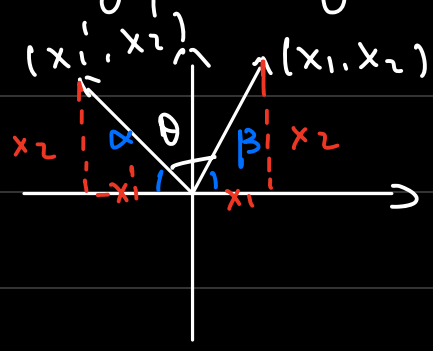
3. Projection onto x-axis

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$



4. Counterclockwise Rotation by an angle θ

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$



proof:

$$\begin{cases} \tan\alpha = -\dfrac{x_2'}{x_1'} \\ \tan\beta = \dfrac{x_2}{x_1} \end{cases}$$

$$\tan(\pi - \alpha - \beta) = \tan\theta$$

$$\Rightarrow -\tan(\alpha+\beta) = \tan\theta$$

$$\Rightarrow -\frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{\sin\theta}{\cos\theta}$$

$$\Rightarrow \frac{\dfrac{x_2'}{x_1'} - \dfrac{x_2}{x_1}}{1 + \dfrac{x_2 x_2'}{x_1 x_1'}} = \frac{\sin\theta}{\cos\theta}$$

$$\Rightarrow \frac{x_2' x_1 - x_2 x_1'}{x_1 x_1' + x_2 x_2'} = \frac{\sin\theta}{\cos\theta}$$

$$\begin{cases} \cos\theta = \dfrac{x_1 x_1' + x_2 x_2'}{x_1^2 + x_2^2} \\ \sin\theta = \dfrac{x_2' x_1 - x_2 x_1'}{x_1^2 + x_2^2} \end{cases}$$

$$\begin{cases} x_1 x_1' + x_2 x_2' = (x_1^2 + x_2^2)\cos\theta \\ x_2' x_1 - x_2 x_1' = (x_1^2 + x_2^2)\sin\theta \end{cases}$$

$$\begin{cases} (x_2^2 + x_1^2)x_2' = (x_2\cos\theta + x_1\sin\theta)(x_1^2 + x_2^2) \\ (x_1^2 + x_2^2)x_1' = (x_1\cos\theta - x_2\sin\theta)(x_1^2 + x_2^2) \end{cases}$$

$$\begin{cases} x_1' = x_1\cos\theta - x_2\sin\theta \\ x_2' = x_2\cos\theta + x_1\sin\theta \end{cases}$$

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Onto and One-to-one Mappings

Def: $T: \mathbb{R}^n \to \mathbb{R}^m$ is **onto** if every output $\vec{b} \in \mathbb{R}^m$ is the image of **at least one** input $\vec{x} \in \mathbb{R}^n$

Def: $T: \mathbb{R}^n \to \mathbb{R}^m$ is **one-to-one** if every output $\vec{b} \in \mathbb{R}^m$ is the image of **at most one** input $\vec{x} \in \mathbb{R}^n$

The: $T: \mathbb{R}^n \to \mathbb{R}^m$ is one-to-one if and only if $T(\vec{x}) = 0$ has only the zero solution / columns of A are linearly independent

Proof: $T(\vec{x_1}) = T(\vec{x_2})$

$\Rightarrow A\vec{x_1} = A\vec{x_2}$

$\Rightarrow A(\vec{x_1} - \vec{x_2}) = 0$

① If the column vector of A linear independent

$A\vec{v} = 0$ only has trivial solution $\vec{v} = \vec{0}$

$\therefore \vec{x_1} - \vec{x_2} = \vec{0}$

$\therefore \vec{x_1} = \vec{x_2}$

This satisfies the defination of one-to-one

② If one-to-one

for $T(\vec{x_1}) = T(\vec{x_2})$,

$\vec{x_1} = \vec{x_2}$

$\therefore \vec{x_1} - \vec{x_2} = \vec{0}$

$\therefore A\vec{v} = 0 \iff A(\vec{x_1} - \vec{x_2}) = \vec{0}$ only has one trivial solution $\vec{0}$

$\therefore$ The column vectors of A are linearly independent

The: $T: \mathbb{R}^n \to \mathbb{R}^m$ is onto if and only if the columns of A span $\mathbb{R}^m$

Proof: ① Onto → Span

If T is onto, for $\forall \vec{b} \in \mathbb{R}^m$, $\exists \vec{x} \in \mathbb{R}^n$

$A\vec{x} = \vec{b}$

expand: $\vec{b} = x_1\vec{a_1} + x_2\vec{a_2} + \cdots + x_n\vec{a_n}$

Satisfies the defination of Span

② Span → Onto

If $\text{Span}\{a_1, a_2, \cdots, a_n\} = \mathbb{R}^m$

$\exists \vec{x} \in \mathbb{R}^n$, for $\forall \vec{b} \in \mathbb{R}^m$

satisfies $x_1\vec{a_1} + x_2\vec{a_2} + \cdots + x_n\vec{a_n} = \vec{b}$

$\Rightarrow A\vec{x} = \vec{b}$

Satisfies the defination of Onto

linear independent

↑

Every Column has pivot : one-to-one

Every Row has pivot : onto

↓

span $\mathbb{R}^m$

$$\begin{matrix} 1 & 0 \\ 0 & -1 \end{matrix}$$